

Day 2

1.

```
> db.products.find()
< {
  _id: ObjectId('6a0fe755e5d07ba06e4a4d14'),
  name: 'iPhone 14',
  category: 'Electronics',
  price: 75000,
  stock: 25,
  brand: 'Apple'
}
{
  _id: ObjectId('6a0fe755e5d07ba06e4a4d15'),
  name: 'Samsung Galaxy S22',
  category: 'Electronics',
  price: 65000,
  stock: 30,
  brand: 'Samsung'
}
```

2.

```
db.products.find().pretty()
{
  _id: ObjectId('6a0fe755e5d07ba06e4a4d14'),
  name: 'iPhone 14',
  category: 'Electronics',
  price: 75000,
  stock: 25,
  brand: 'Apple'
}
{
  _id: ObjectId('6a0fe755e5d07ba06e4a4d15'),
  name: 'Samsung Galaxy S22',
  category: 'Electronics',
  price: 65000,
  stock: 30,
  brand: 'Samsung'
}
```

3.

```
> db.products.findOne({$or: [{name:'Bluetooth Speaker'},{brand:'JBL'}]})
< {
  _id: ObjectId('6a0fe755e5d07ba06e4a4d1d'),
  name: 'Bluetooth Speaker',
  category: 'Electronics',
  price: 2500,
  stock: 60,
  brand: 'JBL'
}
```

4.

```
> db.products.find({category:'Electronics'})
< {
  _id: ObjectId('6a0fe755e5d07ba06e4a4d14'),
  name: 'iPhone 14',
  category: 'Electronics',
  price: 75000,
  stock: 25,
  brand: 'Apple'
}
{
  _id: ObjectId('6a0fe755e5d07ba06e4a4d15'),
  name: 'Samsung Galaxy S22',
  category: 'Electronics',
  price: 65000,
  stock: 30,
  brand: 'Samsung'
}
```

5.

```
> db.products.find({price:{$lt:5000}})
< {
  _id: ObjectId('6a0fe755e5d07ba06e4a4d19'),
  name: 'Adidas T-shirt',
  category: 'Fashion',
  price: 1500,
  stock: 80,
  brand: 'Adidas'
}
{
  _id: ObjectId('6a0fe755e5d07ba06e4a4d1a'),
  name: 'Sony Headphones',
  category: 'Electronics',
  price: 3000,
  stock: 40,
  brand: 'Sony'
}
```

6.

```
> db.products.find({price:{$lte:5000}})
< {
  _id: ObjectId('6a0fe755e5d07ba06e4a4d18'),
  name: 'Nike Shoes',
  category: 'Fashion',
  price: 5000,
  stock: 50,
  brand: 'Nike'
}
{
  _id: ObjectId('6a0fe755e5d07ba06e4a4d19'),
  name: 'Adidas T-shirt',
  category: 'Fashion',
  price: 1500,
  stock: 80,
  brand: 'Adidas'
}
```

7.

```
> db.products.find({price:{$gt:5000}})
< {
  _id: ObjectId('6a0fe755e5d07ba06e4a4d14'),
  name: 'iPhone 14',
  category: 'Electronics',
  price: 75000,
  stock: 25,
  brand: 'Apple'
}
{
  _id: ObjectId('6a0fe755e5d07ba06e4a4d15'),
  name: 'Samsung Galaxy S22',
  category: 'Electronics',
  price: 65000,
  stock: 30,
  brand: 'Samsung'
}
```

8.

```
> db.products.find({price:{$gte:5000}})
< {
  _id: ObjectId('6a0fe755e5d07ba06e4a4d14'),
  name: 'iPhone 14',
  category: 'Electronics',
  price: 75000,
  stock: 25,
  brand: 'Apple'
}
{
  _id: ObjectId('6a0fe755e5d07ba06e4a4d15'),
  name: 'Samsung Galaxy S22',
  category: 'Electronics',
  price: 65000,
  stock: 30,
  brand: 'Samsung'
}
```

9.

```
> db.products.find({price:{$ne:5000}})
< {
  _id: ObjectId('6a0fe755e5d07ba06e4a4d14'),
  name: 'iPhone 14',
  category: 'Electronics',
  price: 75000,
  stock: 25,
  brand: 'Apple'
}
{
  _id: ObjectId('6a0fe755e5d07ba06e4a4d15'),
  name: 'Samsung Galaxy S22',
  category: 'Electronics',
  price: 65000,
  stock: 30,
  brand: 'Samsung'
}
```

10.

```
> db.products.find({price:{$in:[1000,2000,5000]}})
< {
  _id: ObjectId('6a0fe755e5d07ba06e4a4d18'),
  name: 'Nike Shoes',
  category: 'Fashion',
  price: 5000,
  stock: 50,
  brand: 'Nike'
}
```

11.

```
> db.products.find({price:{$nin:[1000,2000,5000]}})
< {
  _id: ObjectId('6a0fe755e5d07ba06e4a4d14'),
  name: 'iPhone 14',
  category: 'Electronics',
  price: 75000,
  stock: 25,
  brand: 'Apple'
}
{
  _id: ObjectId('6a0fe755e5d07ba06e4a4d15'),
  name: 'Samsung Galaxy S22',
  category: 'Electronics',
  price: 65000,
  stock: 30,
  brand: 'Samsung'
}
```

12.

```
> db.products.find({$and: [{category: 'Electronics'}, {price:5000}]})
<
```

13.

```
> db.products.find({$or: [{category: 'Electronics'}],{brand:'Apple'}})
< {
  _id: ObjectId('6a0fe755e5d07ba06e4a4d14'),
  name: 'iPhone 14',
  category: 'Electronics',
  price: 75000,
  stock: 25,
  brand: 'Apple'
}
{
  _id: ObjectId('6a0fe755e5d07ba06e4a4d15'),
  name: 'Samsung Galaxy S22',
  category: 'Electronics',
  price: 65000,
  stock: 30,
  brand: 'Samsung'
}
```

14.

```
> db.products.find({$or: [{price:{$gt:5000}}],{brand:'Samsung'}})
< {
  _id: ObjectId('6a0fe755e5d07ba06e4a4d14'),
  name: 'iPhone 14',
  category: 'Electronics',
  price: 75000,
  stock: 25,
  brand: 'Apple'
}
{
  _id: ObjectId('6a0fe755e5d07ba06e4a4d15'),
  name: 'Samsung Galaxy S22',
  category: 'Electronics',
  price: 65000,
  stock: 30,
  brand: 'Samsung'
}
```

15.

```
> db.products.find({$and: [{price:{$gt:5000}},{price:{$lt:20000}}]})
< {
  _id: ObjectId('6a0fe755e5d07ba06e4a4d1c'),
  name: 'Office Chair',
  category: 'Furniture',
  price: 7000,
  stock: 20,
  brand: 'Godrej'
}
```

16

```
> db.products.find({price:{$not:{$gt:5000}}})
< {
  _id: ObjectId('6a0fe755e5d07ba06e4a4d18'),
  name: 'Nike Shoes',
  category: 'Fashion',
  price: 5000,
  stock: 50,
  brand: 'Nike'
}
{
  _id: ObjectId('6a0fe755e5d07ba06e4a4d19'),
  name: 'Adidas T-shirt',
  category: 'Fashion',
  price: 1500,
  stock: 80,
  brand: 'Adidas'
}
```

Update operations

1.

```
> db.employees.updateOne({_id:ObjectId("6a0fe755e5d07ba06e4a4d15")}, {$set: {price:100000}})
```

2.

```
db.products.updateMany({ price: { $lt: 5000 } }, { $set: { stock: 100 } })
```

3.

```
> db.products.updateOne(
  { category: "Computers" },
  { $set: { price: 75000 } }
)
```

4

```
> db.products.updateMany(
  { category: "Electronics" },
  { $set: { stock: 50 } }
)
```

5.

```
> db.products.findOneAndReplace(
  { _id: ObjectId("6a0fe755e5d07ba06e4a4d1b") },
  {
    name: "Asus laptop",
    category: "Laptop",
    price: 79000,
    stock: 20
  }
)
< {
  _id: ObjectId('6a0fe755e5d07ba06e4a4d1b'),
  name: 'Asus laptop',
  category: 'Laptop',
  price: 79000,
  stock: 20
}
```


Delete Operations

1.

```
> db.products.deleteOne({ name: "Asus laptop" });  
< {  
  acknowledged: true,  
  deletedCount: 1  
}
```

2.

```
> db.products.deleteOne({ price: { $lt: 2000 } });  
< {  
  acknowledged: true,  
  deletedCount: 1  
}
```

3.

```
> db.products.deleteMany({});  
< {  
  acknowledged: true,  
  deletedCount: 8  
}
```

Projection and Filtering

1.

```
> db.products.find({}, { name: 1, price: 1 })
< {
  _id: ObjectId('6a0ff876e5d07ba06e4a4d1e'),
  name: 'iPhone 14',
  price: 75000
}
{
  _id: ObjectId('6a0ff876e5d07ba06e4a4d1f'),
  name: 'Samsung Galaxy S22',
  price: 65000
}
```

2.

```
> db.products.find({}, { name: 1, price: 1, _id: 0 })
< {
  name: 'iPhone 14',
  price: 75000
}
{
  name: 'Samsung Galaxy S22',
  price: 65000
}
```

Limit and Pagination

1.

```
> db.products.find().limit(5)
< {
  _id: ObjectId('6a0ff876e5d07ba06e4a4d1e'),
  name: 'iPhone 14',
  category: 'Electronics',
  price: 75000,
  stock: 25,
  brand: 'Apple'
}
```

2.

`db.products.find().skip(5).limit(5)`

Sorting

1.

`db.products.find().sort({ price: 1 })`

2.

`db.products.find().sort({ price: -1 })`